

Conduction Lead Design and Optimization for Cryogenic Characterization of GaN HEMTs

Tian Qiu
Electrical, Computer, and System
Engineering
Rensselaer Polytechnic Institute
Troy, USA
qiu2@rpi.edu

Zheyu Zhang
Electrical, Computer, and System
Engineering
Rensselaer Polytechnic Institute
Troy, USA
zhangz49@rpi.edu

Purushottam Khadka
Electrical, Computer, and System
Engineering
Rensselaer Polytechnic Institute
Troy, USA
khadkp@rpi.edu

Abstract—Gallium Nitride (GaN) HEMTs present superior characteristics at cryogenic temperatures and becomes promising candidates for the lunar mission under the extreme cold environment down to 50 K. Extensive studies have been performed to characterize GaN HEMTs using cryogenic chamber with Liquid Nitrogen (LN₂) down to 77 K, while few research focuses on cryocooler with Liquid Helium (LHe) with limited trade study of cryogenic cabling between electrical resistance and heat flow. This paper proposes a methodology to optimize lead design for cryogenic device characterization. First, the temperature evolution along the conduction lead during the pulse test is modeled with reasonable assumptions, finding that Joule heating effect is basically negligible. Then, the thermal load and load-line effect are analyzed to introduce the design constraints, by which the optimization is performed to achieve desirable voltage drop and thermal load. Finally, the proposed methodology is validated by several design examples, and the I-V characterization of a GaN HEMT is demonstrated at 50 K.

Keywords—Cryogenic characterization, lead design, load line effect, thermal load

I. INTRODUCTION

Wide-bandgap (WBG) devices exhibited promising advantages over Si counterparts, including the high breakdown voltage, high switching speed and low conduction loss [1]. GaN high electron mobility transistors (HEMTs) are a promising candidate in cryogenic applications [2] and can potentially be utilized for the space power on the lunar surface where the part would experience extremely cold environment down to 50 K [3]. Therefore, the characterization of GaN devices at cryogenic conditions necessitates.

Cryogenic characterization for GaN has been conducted extensively using LN₂ at 77 K. However, few exploration has been conducted at lower temperature down to 50 K. One of the limitations is due to the thermal load imposed in the cooling system, which makes it difficult to reach the expected temperature [4]. Generally, there are three ways of heat transfer to introduce thermal load: conduction, convection and radiation. Considering the conduction leads with one end at room temperature and the other at cryogenic temperature, heat flow will be conducted into the cooling system, imposing undesired thermal load. Thus, the design of leads should be engineered in terms of thermal behavior. On the other hand, the load line effect, as shown in Fig. 1, is unwanted from the perspective of device characterization. Because of the efficient current conduction of GaN HEMTs, this issue becomes crucial since excessive voltage drop induced from large testing current on the leads limits the effective testing voltage applied across the device under test (DUT). Hence, conduction lead resistance should be engineered as well to alleviate this effect.

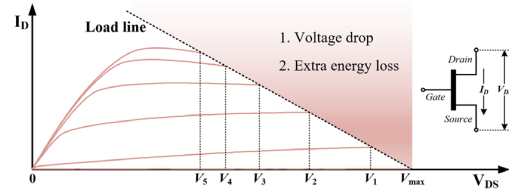


Fig. 1 Load line effect.

The common way of calculating thermal load is to solve the equation of Fourier's conduction law. In terms of numerical solution, Ref. [5] proposed a generic model with discrete expression considering Joule heating effect, such that the optimal length-cross-section area ratio of the lead at specific current can be found. Based on this model, Ref. [6] analyzed the temperature evolution and thermal load of the conduction lead with non-optimum design. As for analytical solution, Ref. [7] proposed analytical equations of thermal load at low- and high-current scenarios with different approximation models to optimize the lead design, which is applicable to arbitrary current conduction. Ref. [8] studied the optimization in the temperature range from 77 K to 300 K, where the thermal conductivity is assumed to be a constant, which simplifies the model. However, most of the existing works mainly aim at thermal load analysis and optimization from the perspective of cryogenic engineering, while the voltage drop concern due to device characterization is not within consideration.

In addition, the conduction lead thermal analysis for semiconductor device characterization has not been studied in detail. A common approach for device characterization is pulse-based test. Taking I-V characterization as an example, the curve tracer generates a sequence of gate control and drain current pulses with limited pulse width, which enables the testing of a broad range of currents while minimizing self-heating effects. With such premise, dedicated approximation could be proposed to simplify the solution of thermal load and electrical resistance derived from Fourier conduction law.

To address the aforementioned issues, a conduction lead design for cryogenic device characterization is proposed in this paper for the design optimization considering the trade-off between thermal load and load line effect. In Section II, the temperature evolution along the lead during device characterization is analyzed. In Section III, the load line effect in characterization test is analyzed and allowable lead electrical resistance is thereby derived. In Section IV, the lead design optimization based on thermal load and electrical resistance is performed. In Section V, the design is validated experimentally at cryogenic temperatures, and I-V characterization of a GaN HEMT is demonstrated at 50 K.

NASA, 22-22-ECF-0025, under award No 80NSSC24K0179

II. THERMAL BEHAVIOR OF CRYOGENIC LEADS

A. Cryocooler setup for device characterization

As shown in Fig. 2, the basic components in the cryocooler implementation comprise 1) vacuum enclosure with sealed connectors, 2) sample holder made of copper and 3) cooling station. Usually, a radiation shield made of aluminum is adopted as well to mitigate the heat radiation from the surrounding. When performing the characterization, the device under test (DUT) is fixed on the sample holder, which is mounted on the cooling station, and is connected to the sealed connector by conduction leads. To minimize the loop area as well as to improve sustainability against EMI, the leads are made as twisted pairs.

During the operation, an external vacuum pump functions to create vacuum environment inside the cryocooler such that heat convection would be restrained, and heat radiation can be also alleviated by radiation shielding. However, due to the existence of conduction leads that act as a thermal conduction path, there is inevitably thermal load imposed on the cryogenic system, which may limit the cooling capacity. If thermal load is too severe, the targeting temperature will be unachievable. Hence, one objective in this paper is to design and evaluate the thermal load to be in an appropriate range.

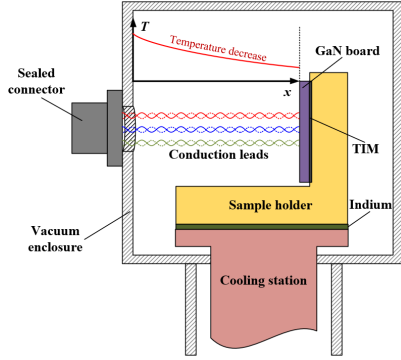


Fig. 2 Cryocooler implementation

B. Temperature evolution modeling

The physical properties used in electrothermal analysis are shown in Fig. 3. The electrical resistivity ρ and specific heat C_{SH} of copper ascend with temperature increase, while thermal conductivity k varies non-monolithically with temperature. At thermal transient state, as temperature changes over time, ρ , C_{SH} and k act as a function of time, while at thermal steady state, they are dependent on only temperature distribution along the conduction lead.

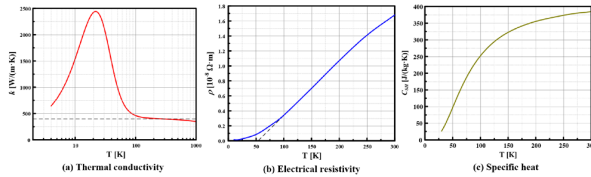


Fig. 3 Thermal conductivity, electrical resistivity and specific heat of copper [9], [10].

Considering the vacuum environment and radiation shield in the cryogenic apparatus, the heat convection and radiation are well mitigated. Thus, the temperature variation along the lead solely obeys Fourier's conduction law as follows:

$$\frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \rho \left(\frac{I}{A} \right)^2 = \lambda C_{SH} \frac{\partial T}{\partial t} \quad (1)$$

Here, A is the cross-section area of the lead, λ is the density of lead material, and C_{SH} is the specific heat.

In device characterization, the pulse-based approach is generally used such that the self-heating effect can be well mitigated. The testing current can be approximated as a piecewise function as follows:

$$I = \begin{cases} I_D, & 0 \leq t < t_m \\ 0, & t_m \leq t < t_p \end{cases} \quad (2)$$

where t_m is the pulse width and t_p is the period in the test. Usually, t_m is 10 – 100 μ s and t_p is 1 – 10 s, indicating the duty ratio ($D = t_m/t_p$) is 10^{-5} – 10^{-4} such that the self-heating effect will be refrained. To obtain the average temperature during the test, Eq. (1) is integrated over t_p as follows:

$$\frac{\partial}{\partial x} \int_0^{t_p} k \frac{\partial T}{\partial x} dt + \frac{1}{A^2} \int_0^{t_p} \rho I^2 dt = \int_{T(x,0)}^{T(x,t_p)} \lambda C_{SH} dT \quad (3)$$

Since ρ is dependent on T , which increases due to Joule heating effect at $0 \leq t < t_m$ and decreases at $t_m \leq t < t_p$, ρ is thereby a function with t , making it difficult to analyze the integral $\int_0^{t_p} \rho I^2 dt$. To estimate the temperature increment induced by Joule heating effect, the lead is regarded as a series of small elements with the length of dx , the temperature variation on which is almost constant, allowing us to approximate k and ρ to be independent of x . Hence, the maximum temperature increment induced by I_D at $t = t_m$ is calculated from Eq. (1):

$$\Delta T = \frac{\rho t_m}{\lambda C_{SH}} \left(\frac{I_D}{A} \right)^2 \quad (4)$$

In Eq. (4), ρ and C_{SH} are functions of temperature, while A and λ almost remain unchanged with temperature. Using Eq. (4), the temperature increment is estimated. For example, if a 20 AWG copper wire conducts a pulse current $I_D = 100$ A, $t_m = 100$ μ s, the maximum temperature increment is $\Delta T = 0.02$ K, which is negligible. Therefore, it is reasonable to assume that ρ , C_{SH} and k remain unchanged during the thermal transient process. In addition, considering the fact that $T(x, 0)$ has to equal to $T(x, t_p)$ at thermal steady state, the right-hand side of Eq. (3) is naturally zero. Accordingly, Eq. (3) can be written as follows:

$$\frac{\partial}{\partial x} \left(k \frac{\partial T_{ave}}{\partial x} \right) + \rho \left(\frac{I_{RMS}}{A} \right)^2 = 0 \quad (5)$$

where

$$T_{ave} = \frac{1}{t_p} \int_0^{t_p} T dt$$

$$I_{RMS} = I_D \sqrt{D} \quad (6)$$

Here, T_{ave} is the time-averaged temperature along the lead. By solving Eq. (5), the average temperature distribution along the conduction lead at thermal steady state is obtained. The solution of Eq. (5) is:

$$T_{ave} = C_1 x + C_2 - \left(\frac{I_{RMS}}{A} \right)^2 \int \frac{1}{k} \left(\int \rho dx \right) dx \quad (7)$$

where C_1 and C_2 are constants from integral. The integra term in Eq. (7) reflects Joule heating contribution to T_{ave} . If I_{RMS} is small enough, T_{ave} linearly changes along the lead, indicating the heat conduction induced from temperature gradient makes up the thermal load.

To estimate Joule heating effect of conduction leads in cryogenic characterization, Eq. (8) is proposed as follows:

$$\left(\frac{I_{RMS}}{A}\right)^2 \int \frac{1}{k} \left(\int \rho dx \right) dx < \frac{\rho_{295K}}{2k_{295K}} \left(\frac{I_{RMS}L}{A} \right)^2 \quad (8)$$

Here, L is the length of conduction leads. Taking EPC2302 as an example, the current rating of which is 101 A. Assuming $D = 2 \times 10^{-5}$ (assume $t_m = 40 \mu s$ and $t_p = 2 s$ in (2)) and the conduction lead used in cryogenic characterization is 20 AWG / 0.2 m, Joule heating effect on lead temperature is < 0.6 K, which is negligible compared to T_{ave} . Therefore, by appropriate conduction lead design, the thermal impact from the testing current can be deemphasized.

C. Thermal load

The thermal load is defined as the heat flow into the cryogenic apparatus, which is $Q_{lead} = -kAdT/dx$ [5]. With regardless of Joule heating effect, the thermal load at thermal steady state can be derived from Eq. (5) as follows:

$$Q_{lead} = \frac{1}{Z} \int_{T_L}^{T_H} k dT \quad (9)$$

where Z is the length-area (L-A) ratio ($Z = L/A$), T_H and T_L are respectively the warm-end and cold-end temperature of the lead. Using COMSOL simulation, Eq. (9) is well validated at different T_L with T_H fixed at room temperature, as shown in Fig. 4, which indicates a good correlation. Based on Eq. (9), the thermal load is reciprocal to the L-A ratio, indicating that designing longer conduction leads can alleviate the thermal load to the cryogenic system.

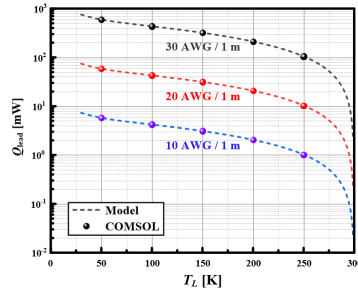


Fig. 4 Validation of thermal load calculation in simulation.

III. LOAD LINE EFFECT IN DEVICE CHARACTERIZATION

A. Resistance constraint

Taking static characterization as an example, as depicted in Fig. 5, there are basically three types of conduction leads: forcing, sensing and gate control leads [12]. The forcing lead pair apply testing current to the DUT, which inevitably generates a voltage drop on the leads, making DUT's drain-to-source voltage (V_{DS}) actually smaller than curve tracer's output (V_{CT}). The conduction lead resistance, together with the output resistance of curve tracer contribute to the load line effect, which limits the testing range and induce extra energy loss during the test.

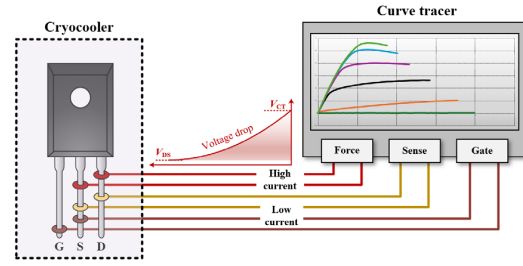


Fig. 5 Conduction leads in static characterization

To characterize the device for its full operation range including the linear region and saturation region, the maximum voltage drop along the lead should be estimated. A piecewise function is used to analyze V_{DS} as follows:

$$V_{DS} = \begin{cases} \frac{V_{CT}R_{on}}{R_{on} + R_{ex} + R_{lead}}, & I_D < I_{sat} \\ V_{CT} - I_{sat}(R_{ex} + R_{lead}), & I_D = I_{sat} \end{cases} \quad (10)$$

Here, R_{on} is the on-state resistance, which is a function of gate-to-source voltage (V_{GS}), R_{ex} is the external resistance that sits outside the cryocooler, including output resistance of the curve tracer, resistance of external cable and PCB traces, R_{lead} is the resistance of the cryogenic lead (i.e., the internal lead), and I_{sat} is the saturation current which is also a function of V_{GS} . Suppose the voltage ranged in the curve tracer setup is $0 - V_{CT}$, to make sure the I-V characteristics can get into saturation, the maximum allowable voltage drop ΔV_{max} should meet the requirement as follows:

$$\Delta V_{max} = I_{sat}(R_{ex} + R_{lead}) < V_{CT} - R_{on}I_{sat} \quad (11)$$

Here, I_{sat} and R_{on} are at cryogenic condition. Solving Eq. (11), the design constraint for R_{lead} is obtained as follows:

$$R_{lead} < \frac{V_{CT}}{I_{sat}} - (R_{on} + R_{ex}) \quad (12)$$

Based on the preliminary test results (will be later shown in Section V), however, I_{sat} of GaN HEMTs at cryogenic conditions will be significantly increased, in which case Eq. (11) may not hold. In addition, I_{sat} at cryogenic temperature is unknown before the test, making the resistance design at cryogenic conditions is difficult since only I_{sat} at room temperature is available in the datasheet. Thus, an alternative design constraint is thereby proposed as follows:

$$R_{lead} < \frac{R_{ex}}{10} \quad (13)$$

By using Eqs. (12) and (13), the designed cryogenic conduction leads will have limited contribution to the load line effect during characterization.

B. Cryogenic resistance

Considering the analysis in Section II, the electrical resistivity ρ varies as a function of temperature along the conduction lead. For each element with the length of dx on the lead, its resistance dR_{lead} is obtained as follows:

$$dR_{lead} = \frac{\rho dT}{A} \left(\frac{dx}{dT} \right) = -\frac{\rho k dT}{Q_{lead}} \quad (14)$$

Integrating Eq. (14) over x , R_{lead} is obtained in Eq. (15).

$$R_{\text{lead}} = \frac{Z \int_{T_L}^{T_H} \rho k dT}{\int_{T_L}^{T_H} k dT} \quad (15)$$

The validation of Eq. (15) is shown in Fig. 6. The calculation result is well in line with the finite-element analysis results. In Eq. (15), R_{lead} is proportional to the L-A ratio, which is on the contrary with Q_{lead} . Hence, if the lead resistance is designed to be smaller, its thermal load to the cryogenic system will be more severe, and vice versa, indicating a trade-off should be made for the optimal design.

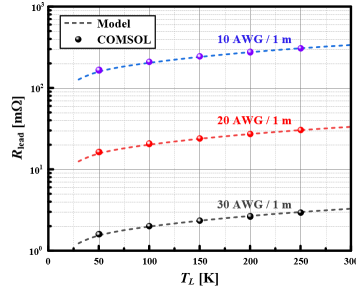


Fig. 6 Validation of electrical resistance calculation in simulation.

IV. DESIGN OPTIMIZATION OF CRYOGENIC LEADS

The lead design and optimization should account for heat transfer as well as load line effect. Given the fact that the design for low thermal load is conflicting with that for low electrical resistance, appropriate design strategies should be proposed based on specific functions of conduction leads. In device characterization as shown in Fig. 5, only forcing leads conduct high current, and gate control and sensing leads are used for low current conduction. In terms of the number of leads, low current leads make up 67% in the system, which probably will introduce more thermal load than high current leads do. Therefore, reducing thermal load is the main objective of sensing and gate control lead design, while reducing electrical resistance is the priority of forcing lead design.

In cryogenic apparatus, phosphor bronze wires are generally used for low current conduction due to its excellent thermal behavior [9]. The thermal conductivity of phosphor bronze is only 15%–20% of copper's [10], indicating a much lower heat flow from warm end to cold end. Despite its higher electrical resistivity, as sensing or gate control wires, the lead resistance is not a design concern. On the other hand, copper wires are usually used for high current conduction because of the low electrical resistivity. However, to ensure the total thermal load sits within the cooling capacity of the system, the forcing leads should be designed to mitigate the high thermal load due to the high thermal conductivity of copper.

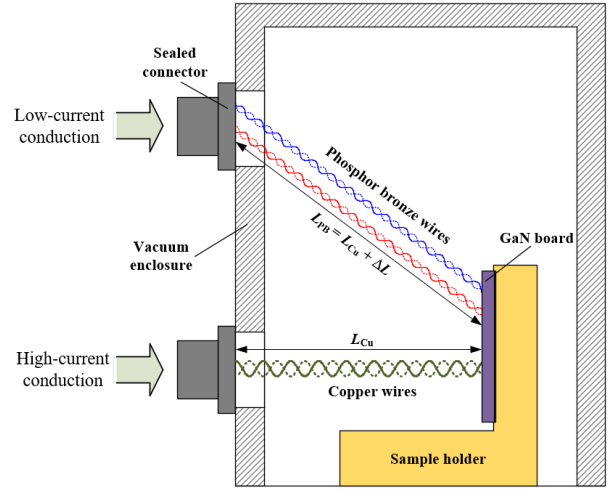


Fig. 7 Lead design within the cryocooler.

Hence, the design of high- and low-current conduction leads should be considered separately. Overall, the lead length to some degree depends on the internal structure of the cryocooler, as shown in Fig. 7. Once the length is determined, the cross-section area is then determined by current rating and constraints in Eqs. (9), (12) and (13). According to Fig. 7, the length correlation between high-current copper leads and low-current phosphor bronze leads should meet the equation as follows:

$$L_{\text{PB}} = L_{\text{Cu}} + \Delta L \quad (16)$$

where L_{Cu} and L_{PB} are respectively the length of copper and phosphor bronze leads, and ΔL is the length difference. The length design in Eq. (16) can be achieved by minimizing the objective function in Eq. (17), where A_{Cu} and A_{PB} are the cross-section area of copper and phosphor bronze leads, and Z_{Cu} and Z_{PB} are the L-A ratio of copper and phosphor bronze.

$$f_{\text{obj}} = |A_{\text{PB}} Z_{\text{PB}} - A_{\text{Cu}} Z_{\text{Cu}} - \Delta L| \quad (17)$$

Based on Sections II and III, the design constraints regarding thermal load and electrical resistance are obtained as follows:

$$\begin{cases} \frac{N_{\text{Cu}} \int_{T_L}^{T_H} k_{\text{Cu}} dT}{Z_{\text{Cu}}} + \frac{N_{\text{PB}} \int_{T_L}^{T_H} k_{\text{PB}} dT}{Z_{\text{PB}}} < Q_{\text{max}} \\ \frac{Z_{\text{Cu}} \int_{T_L}^{T_H} \rho_{\text{Cu}} k_{\text{Cu}} dT}{\int_{T_L}^{T_H} k_{\text{Cu}} dT} < R_{\text{max}} \end{cases} \quad (18)$$

where

$$R_{\text{max}} = \frac{1}{2} \max \left[\frac{V_{\text{CT}}}{I_{\text{sat}}} - (R_{\text{on}} + R_{\text{ex}}), \frac{R_{\text{ex}}}{10} \right] \quad (19)$$

Here, N_{Cu} and N_{PB} is the number of copper and phosphor bronze leads, Q_{max} is the maximum thermal capacity of cryocooler and R_{max} is the maximum allowable resistance for a single high-current conduction lead, which is determined by Eqs. (12) and (13). Given that low-current leads have much smaller wire gauge than high-current leads do, Z_{PB} is usually way larger than Z_{Cu} . In addition, k_{PB} is much smaller than k_{Cu} . Therefore, the thermal load of copper wires is dominant

in Eq. (18), indicating that the high-current lead optimization is critical to the design trade-off.

With Eq. (18), the solution area of Z_{Cu} and Z_{PB} that meet the requirements of thermal load and electrical resistance is determined, as shown in Fig. 8. The solutions that meet Eq. (16) are on the blue dash line within the solution area in Fig. 8. The optimal design result can be found around the midpoint of the section in the solution area to remain sufficient and equal margins from boundaries of the solution area.

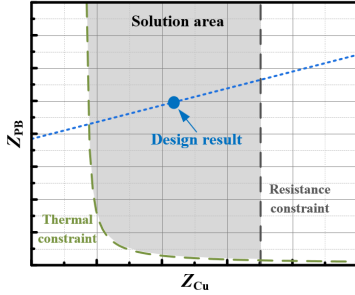


Fig. 8 Optimization design.

Overall, the lead design optimization is depicted by a flow chart in Fig. 9. Firstly, based on the number of device within the cryocooler, N_{Cu} and N_{PB} are determined. Then, appropriate values of L_{Cu} and L_{PB} are selected according to the internal structure of the vacuum chamber. Based on N_{Cu} and N_{PB} , the solution area for all Z_{Cu} and Z_{PB} can be determined using Eq. (18). By finding the midpoint of the overlapped section of Eq. (17) with the solution area, the optimal design is obtained. Finally, use Eq. (17) to back calculate A_{Cu} and A_{PB} with the optimal Z_{Cu} and Z_{PB} . If A_{Cu} cannot meet the current conduction rating, either the values of L_{Cu} and L_{PB} should be redesigned or to find another Z_{Cu} and Z_{PB} from the solution area.

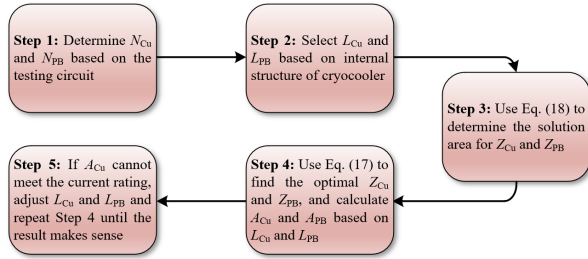


Fig. 9 Cryogenic lead design flow chart.

V. CRYOGENIC TEST

A. Cryogenic apparatus

Based on cryocooler CTI-Cryogenics M-22, we design dedicated accessories including vacuum enclosure, sample holder and radiation shield to build the cryogenic apparatus for device characterization, as shown in Fig. 10. The GaN HEMT board is mounted on the sample holder, and thermal grease is applied to improve heat transfer efficiency. The sample holder sits on the cooling station of the cryocooler, with a indium foil in between as a thermal interface. A radiation shield made of polished aluminum is mounted on the cooling station and surrounding the sample to mitigate heat radiation from the vacuum enclosure. On top of that, the designed cryogenic conduction leads connect to the sample board through a interface board.

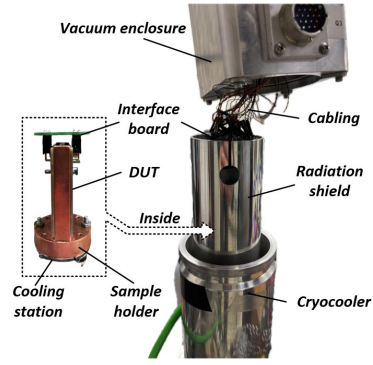


Fig. 10 Cryocooler accessories design, including vacuum chamber, sample holder and cabling.

In the operation of the cryogenic apparatus, an external vacuum pump reduces the pressure to be ~ 100 mTorr, and then the compressor initiates the thermal cycling of LHe_2 inside the cryocooler, which consistently cools down the sample through the cooling station and sample holder, until the temperature reaches the set point in of the temperature controller. Then, the controller will powers up the heater in the cryocooler to balance its cooling capacity to thereby stabilize the temperature at the desired value.

B. Design validation

Since two basic assumptions are made in Sections II and III for modeling Q_{lead} and R_{lead} : 1) the cryocooler has ideal vacuum inside, by which heat convection with environment is negligible and 2) the impact of heat radiation is not considered. In terms of the designed cryocooler accessories, the vacuum enclosure and radiation shield are respectively used to alleviate the heat convection and radiation to thereby hold the assumptions. To examine the performance of the design as well as to validate the conduction lead modeling, Q_{lead} and R_{lead} should be evaluated. Thus, Eqs. (9) and (15) are respectively to be validated as follows.

Case 1: Thermal load design

Direct measurement of Q_{lead} is difficult, but it is possible to use estimate it by the output power of heater. At thermal steady state, the cooling capacity of cryocooler is balanced by the total thermal load including heater output Q_{heater} , environmental heat convection and radiation, and Q_{lead} . Therefore, Q_{lead} can be approximated as the heater output difference with and without conduction leads connected to the sample holder, as expressed in Eq. (20).

$$Q_{lead} \approx (n_{w/o}^2 - n_w^2) I_{he}^2 R_{he} \quad (20)$$

Here, n_w and $n_{w/o}$ are respectively the operating current percentage of heater with and without cryogenic leads, I_{he} is heater's maximum operating current ($I_{he} = 0.3$ A) and R_{he} is heater resistance ($R_{he} = 25 \Omega$)

To validate Eq. (9), 12 copper wires (26 AWG) with the length of 15 – 17 cm are connected to the sample holder at $T_L = 50$ K and $T_H = 295$ K. Using Eq. (9), the estimated Q_{lead} is around 1 W. The experimental result is shown in Fig. 11, where $n_w = 45\%$ and $n_{w/o} = 75\%$, indicating the actual Q_{lead} is about 0.81 W, which is fair close to the calculation result. The error could mainly originate from the measurement itself, since the resolution of heater power

indicator is 0.25 W. Therefore, Eq. (9) is well validated at cryogenic temperature.



Fig. 11 Heater output current percentage at 50 K with and without conduction leads is respectively 45% and 75% of the maximum current.

Case 2: Electrical resistance design

To validate Eq. (15), a single copper wire is used as the sample. The testing circuit is shown in Fig. 12, including a digital power supply (Keithley 2230-30-1), a high resolution multimeter (Keithley DAQ6510) and a resistor (1 Ω / 10 W). The 4-wire Kelvin connection is used to measure sample's resistance. The sample is twisted and tied to the sample holder to control the cold-end temperature, with its warm-end temperature fixed at 295 K.

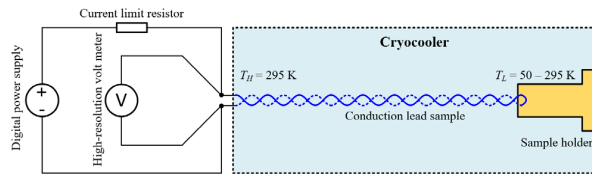


Fig. 12 Conduction lead resistance measurement.

In order to obtain the exact L-A ratio, the sample's resistance is first measured at room temperature (i.e., $T_H = T_L = 295$ K) such that the L-A ratio is back calculated ($Z_{\text{sample}} = 2.37 \times 10^6$). In Fig. 13, the experiment results are plotted together with the error. The calculation results are in line with the experiment, with the maximum error of 6.5% appearing at 50 K. As T_L rises, the error gradually decreases, a possible explain of which is that the error is mainly induced from heat radiation. At low temperature, the influence of radiation is more significant, thus leading to a larger error in the calculation. Overall, Eq. (15) is well validated by the experiment. Therefore, the design constraints proposed in Eq. (18) makes sense.

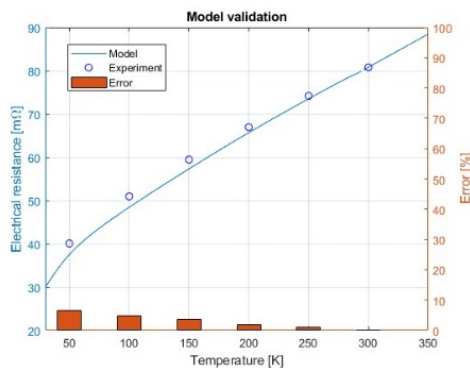


Fig. 13 Sample's resistance versus the cold-end temperature from 50 K to 295 K, with the warm-end temperature fixed at 295 K.

Case 3: Cryogenic device characterization

Based on the design requirements in Table I, conduction leads are design and shown in Table II. As shown in the results, phosphor bronze leads have almost no contribution to

the thermal load compared with copper leads, because on the one hand the L-A ratio is very small and on the other hand the thermal conductivity is very small as well. Hence, the design of copper leads for high-current conduction is the major concern.

TABLE I. DESIGN REQUIREMENTS

N_{Cu}	N_{PB}	V_{CT} [V]	R_{max} [m Ω]	Q_{max} [W]
6	16	20	12.5	2

TABLE II. DESIGN RESULTS @ 50 K

Material	AWG	L [cm]	R [m Ω]	Q [W]
Copper	28 ($\times 4$)	20 – 25	5 – 6	0.9 – 1.1
Phosphor bronze	32	20 – 25	N/A	< 0.03

With the implementation of designed conduction leads, a GaN HEMT (GS-065-018-2-L) is characterized at both room 300 K and 50 K, as shown in Fig. 14. As temperature decrease from 300 K to 50 K, due to the significantly increased drain current in the test, the load line effect is much more severe, making the saturation region unavailable and increasing power loss. According to the design results in Table 2, the resistance of conduction leads inside the cryocooler is sufficiently small, indicating R_{ex} is primary cause of load line effect. Therefore, it is also necessary to minimize the length of the external cable to further mitigate the load line effect especially at cryogenic conditions.

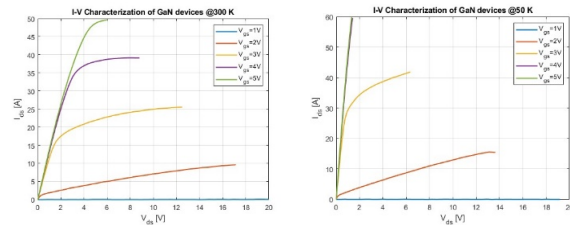


Fig. 14 I-V characterization @300 K and 50 K of GS-065-018-2-L.

VI. CONCLUSIONS

The paper proposes a design methodology for conduction leads utilized in cryogenic device characterization, which aims at alleviating both thermal load of the cryogenic system and load line effect of the curve tracer. Analysis shows that in the pulse-based device characterization, the thermal impact of Joule heating effect on the lead's temperature is negligible, by which the lead thermal load can be approximated as solely from heat conduction. Accordingly, analytical expressions of thermal load and electrical resistance of cryogenic conduction leads are derived, which are well validated by simulation, indicating that reducing thermal load and electrical resistance conflicts with each other. By design optimization, a trade-off between thermal load and electrical resistance is achieved, the key to which is the design of high-current conduction leads. Finally, the proposed design is validated by experiments, where the calculation is basically in line with the measurement results. At cryogenic conditions, the load line effect primarily depends on R_{ex} and becomes severe, making it necessary to minimize the length of external cables.

ACKNOWLEDGMENT

This material is based on work supported by NASA under Award No. 80NSSC24K0179. The authors would also like to

thank Mr. Richard C. Oeftering, Dr. Nicholas R. Ugucini, Dr. Brian A. Holler, and Mr. Frederick W. Van Keuls from NASA Glenn Research Center for their contributions to the design of the cryocooler platform. Additionally, we extend our gratitude to Prof. Ishwara Bhat from Rensselaer Polytechnic Institute for his assistance with the operation and maintenance of the cryogenic apparatus.

REFERENCES

- [1] T. P. Chow et al., "Smart Power Devices and ICs Using GaAs and Wide and Extreme Bandgap Semiconductors," in *IEEE Transactions on Electron Devices*, vol. 64, no. 3, pp. 856-873.
- [2] M. u. Hassan et al., "Design and Validation of a 20-kVA, Fully Cryogenic, Two-Level GaN-Based Current Source Inverter for Full Electric Aircrafts," in *IEEE Transactions on Transportation Electrification*, vol. 8, no. 4, pp. 4743-4759, Dec. 2022.
- [3] Oeftering, Richard C. "Power Hibernation for Low-Cost Solar Powered Lunar Missions." In *Future In-Space Operations (FISO) Telecon Seminar Presentations*. 2023.
- [4] Z. Zhang et al., "Characterization of Wide Bandgap Semiconductor Devices for Cryogenically-Cooled Power Electronics in Aircraft Applications," 2018 AIAA/IEEE EATS, Cincinnati, OH, USA, 2018, pp. 1-8.
- [5] McFee, Richard. "Optimum input leads for cryogenic apparatus." *Review of Scientific Instruments* 30.2 (1959): 98-102.
- [6] Kar, Soumen, et al. "Performance test and thermal analysis of conduction-cooled optimized current leads at non-optimum operation." *AIP Conference Proceedings*. Vol. 1434. No. 1. American Institute of Physics, 2012.
- [7] Chang, Ho-Myung et al., "Optimization of conduction-cooled current leads with unsteady operating current." *Cryogenics* 49.5 (2009): 210-216.
- [8] Xuan, X. C., et al. "Optimization and thermodynamic understanding of conduction-cooled Peltier current leads." *Cryogenics* 42.2 (2002): 141-145.
- [9] Ekin, Jack. *Experimental techniques for low-temperature measurements: cryostat design, material properties and superconductor critical-current testing*. Oxford university press, 2006.
- [10] *Properties of Copper and Copper Alloys at Cryogenic Temperature* NIST Monograph 177 N.J. Simon, E.S. Drexler, and R.P. Reed February 1992, p 7-23.
- [11] The Engineering ToolBox (2023). Copper - Density, Specific Heat and Thermal Conductivity vs. Temperature. [Online] Available at: https://www.engineeringtoolbox.com/copper-density-specific-heat-thermal-conductivity-vs-temperature-d_2223.html [Accessed 29 Feb. 2024].
- [12] F. Wang, Z. Zhang, and E. A. Jones, *Characterization of Wide Bandgap Power Semiconductor Devices*. London, U.K.: IET Press, 2018, p. 19-2.